

Applied AI and Research Evidence

Tarik Ziya Aykut - selected evidence for applied artificial intelligence, biomedical image analysis, technical implementation, and research contribution.

#	Evidence	Purpose / relevance
1	Applied AI / PeerJ research summary	Summarizes cilia detection and motion phenotyping project, applicant role, metrics, and supervisor contact.
2	PeerJ manuscript title page	Shows manuscript title and author list including the applicant.
3	Author contribution page	Shows contribution statement that the applicant performed experiments, analyzed data, and reviewed drafts.
4	Yetgen certificate	Additional training and development evidence.

Note: This is a focused research/special-achievement package. Debate and voluntary leadership are kept in a separate package.

Applied AI / PeerJ Research Evidence

Manuscript: Automated Hierarchical Classification of Ciliary Motion Phenotypes Using Optical Flow and Machine Learning for Primary Ciliary Dyskinesia

Journal: PeerJ Computer Science - under review

PeerJ manuscript URL: <https://peerj.com/manuscripts/131790/> (may require author/reviewer access)

Applicant role: Co-author / technical implementer. The manuscript lists Tarık Ziya Aykut as an author, and the contribution statement states that he performed experiments, analyzed data, authored or reviewed drafts, and approved the final draft.

Academic supervisor/reference contact: Asst. Prof. Dr. Mehmet Serkan Apaydın -
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Full technical report link from CV: Cilia Project Report

Key technical evidence from the project report:

- Prepared by Tarık Ziya Aykut and İsmail Kılıç as technical implementers, under PI Elif Sever and academic supervisor Asst. Prof. Dr. Mehmet Serkan Apaydın.
- Built a deep-learning pipeline for detecting and segmenting cilia structures in respiratory epithelial microscopy videos.
- Expanded the annotated dataset from 153 frames to 996 frames, with 796 training and 200 validation images.
- Trained and compared YOLOv8n-seg, YOLOv8s-seg, YOLOv8m-seg, and YOLO11m-seg architectures.
- Best validation model achieved 0.93 AP50; independent validation reported 0.839 precision and 0.836 F1-score.

PeerJ page screenshot provided by applicant:

ABSTRACT

Primary Ciliary Dyskinesia (PCD) is a genetically heterogeneous disorder characterized by impaired ciliary motility. Classical High Speed Video Microscopy (HSVM) literature describes three abnormal ciliary motion phenotypes: immotile, stiff, and circular; yet, quantitative evaluation and automated discrimination of these patterns remain challenging. In this study, we developed and validated a fully automated framework for classifying ciliary motion phenotypes based on public and Marmara University Pediatric Pulmonology HSVM databases. The system integrates YOLO8 based automated segmentation, dense TV-L1 optical flow, vorticity and strain descriptors, and wave-based temporal features within a two-stage hierarchical classifier: stage-1 distinguishes immotile from motile recordings with an accuracy 0.95, and stage-2 differentiates stiff and circular phenotypes. Class imbalance was addressed using SMOTE-based oversampling. With the addition of healthy ciliary motion as a normal class, the four-class model (immotile, normal, stiff, circular) achieved 80–84% accuracy under strict patient-based cross-validation, while the clinically oriented three-class model (immotile, stiff, circular) reached 83–87% accuracy. Immotile recordings were consistently identified with near-perfect recall, whereas stiff and circular phenotypes were limited by patient-level sparsity but improved substantially in video-based evaluation (balanced accuracy ≈0.73). To our knowledge, this represents the first automated system capable of distinguishing multiple impaired ciliary motion phenotypes directly from HSVM recordings. Feature attribution analyses showed that the model relies on physiologically meaningful descriptors, including CBF, magnitude quantiles, vorticity, and angular stability, indicating strong alignment with established biomechanical principles. Overall, this optical flow based hierarchical machine-learning classifier provides an interpretable and extensible foundation for automated HSVM phenotyping and highlights the need for larger multicenter datasets to improve recognition of rare waveform abnormalities in PCD.

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Automated hierarchical classification of ciliary motion phenotypes using optical flow and machine-learning for primary ciliary dyskinesia

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Primary Ciliary Dyskinesia (PCD) is a genetically heterogeneous disorder characterized by impaired ciliary motility. Classical High Speed Video Microscopy (HSVM) literature describes three abnormal ciliary motion phenotypes: immotile, stiff, and circular; yet, quantitative evaluation and automated discrimination of these patterns remain challenging. In this study, we developed and validated a fully automated framework for classifying ciliary motion phenotypes based on public and Marmara University Pediatric Pulmonology HSVM databases. The system integrates YOLO8 based automated segmentation, dense TV-L1 optical flow, vorticity and strain descriptors, and wave-based temporal features within a two-stage hierarchical classifier: stage-1 distinguishes immotile from motile recordings with an accuracy 0.95, and stage-2 differentiates stiff and circular phenotypes. Class imbalance was addressed using SMOTE-based oversampling. With the addition of healthy ciliary motion as a normal class, the four-class model (immotile, normal, stiff, circular) achieved 80–84% accuracy under strict patient-based cross-validation, while the clinically oriented three-class model (immotile, stiff, circular) reached 83–87% accuracy. Immotile recordings were consistently identified with near-perfect recall, whereas stiff and circular phenotypes were limited by patient-level sparsity but improved substantially in video-based evaluation (balanced accuracy ≈ 0.73). To our knowledge, this represents the first automated system capable of distinguishing multiple

The hierarchical design effectively isolates immotile cilia as a simple and highly separable motion state, while enabling fine-grained discrimination among motile subtypes in the second stage. Thereby addressing ciliary motion patterns that remain difficult to distinguish by expert visual inspection alone. Together, these findings show that fine-grained, automated HSVM phenotyping is both feasible and biologically grounded, with remaining limitations driven primarily by the limited representation of rare circular and stiff patterns rather than methodological constraints. In addition to four-class motion phenotype classification, an exploratory video-based five-class analysis further subdivided ciliary motion phenotypes into clinically defined subgroups, demonstrating that optical-flow-derived features capture fine-grained differences within motile cilia patterns. As multi-video and multi-center datasets grow, this framework offers a scalable path toward standardized, objective, and reproducible PCD motion assessment.

Additional Information and Declarations

Competing Interests

The authors declare there are no competing interests.

Author Contributions

Elif Arik Sever conceived and designed the experiments, performed the experiments, analyzed the data, prepared figures and/or tables, authored or reviewed drafts of the article, and approved the final draft.

Tarık Aykut performed the experiments, analyzed the data, authored or reviewed drafts of the article, and approved the final draft.

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CERTIFICATE OF COMPLETION

Tarık Ziya Aykut

Has participated in the 88 hour Yetkin Gençler training program organized with the support of Mehmet Zorlu Foundation and MEF University between September 25, 2021 and December 12, 2021 and was entitled to receive this certificate.

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